

Direct-Printing Hydrogel-Based Platform for Humidity-Driven Dynamic Full-Color Printing and Holography

Chenjie Dai, Zile Li, Zhe Li, Yangyang Shi, Zejing Wang, Shuai Wan, Jiao Tang, Yongquan Zeng, and Zhongyang Li*

Hydrogel materials endow the flat optics platform with active tuning capability, owing to their remarkable humidity-responsive swelling behavior. Despite recent advances in hydrogel-based devices for spectral tuning, their complex patterning processing and limited functionality obstruct them from practical applications. Herein, a single-step direct-printing technique is originally demonstrated with an active hydrogel material platform for realizing unprecedented multi-field full-color display dynamics. Through exploring the dose-induced shrinkage on polyvinyl alcohol, the stepwise hydrogel nanocavities sandwiched by ultrathin metallic films can be directly printed by grayscale e-beam lithography. Due to the tunable structural coloration from hydrogel nanocavity scaling, multi-functionalities are successfully created for optical concealment, dynamic coloring, and dynamic full-color holography. By encoding the cavity-dependent transmission phase into the direct-printed hydrogel platform, it originally enables the projected full-color holographic dynamics in real-time by simply exhalation, beyond the static holographic display or monochromatic holographic switching. The proposed active displays can rapidly respond to the surrounding humidity change at a millisecond-level (< 150 ms). Such a direct-printing strategy for hydrogel nanocavity represents a critical advance toward the unprecedented dynamic full-color display, and suggests promising applications in optical security, gas sensing, multispectral imaging, full-color holography, and next-generation display techniques.

C. Dai, Z. Li, Z. Li, Y. Shi, Z. Wang, S. Wan, J. Tang, Y. Zeng, Z. Li
Electronic Information School
Wuhan University
Wuhan 430072, China
E-mail: zhongyangli@whu.edu.cn

Z. Li, Z. Li
Suzhou Institute of Wuhan University
Suzhou 215123, China

Z. Li, Y. Zeng
Peng Cheng Laboratory
Shenzhen 518055, China

Y. Zeng, Z. Li
Wuhan Institute of Quantum Technology
Wuhan 430206, China

Z. Li
School of Microelectronics
Wuhan University
Wuhan 430072, China

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/adfm.202212053>.

DOI: 10.1002/adfm.202212053

1. Introduction

Flat optics and metasurfaces, as representatives of planar nanophotonics, exhibit brilliant capability of arbitrarily manipulating light field in subwavelength scale,^[1–4] thus promoting various applications for structural color,^[5–8] beam steering,^[9–12] meta-holography,^[13–18] lensing,^[19–21] etc. To endow the post-fabrication tuning ability of the optical devices, several tunable materials such as liquid crystals,^[22–28] phase-change materials,^[29–31] transparent conductive materials,^[32,33] elastomeric polymer,^[34–36] doped semiconductors,^[37,38] and chemically-responsive materials^[39] have been employed to construct the nano-devices. Although a variety of impressive demonstrations have been achieved using those tunable materials, a critical challenge for meta-optics devices is to realize versatile, colorful displays with a practical tuning mechanism, high-speed responsiveness, continuous tuning, simplified fabrication, and large-area optical modulation.

Direct-printing technique is desirable and promising for simplifying the nano-fabrication process, which allows direct

shaping of the material at the nano-/micro-scale using the ultrafast laser,^[40] ultraviolet,^[41] or electron-beam exposure^[7] without additional process. However, the previous direct-printing works mostly exhibit only static functionality including static structural colors, but lack the active regulation capability and mechanisms.

Hydrogel material with its remarkable humidity-responsive swelling properties could enable optical modulation by utilizing its geometric change, thus providing a potential route to meet those challenges. Recently, several typical hydrogels such as chitosan and polyvinyl alcohol (PVA) have been exploited in spectral tuning,^[42–44] but suffer complex patterning processes or limited functionality. For instance, chitosan-based Fabry-Perot (FP) etalon has been demonstrated to realize ultrafast humidity-responsive colorimetric sensors and colorful printing display,^[43] however, its various chitosan cavity thickness cannot be directly fabricated but is remedied by growing on multistep silica layer, which requires a series of repeated photolithography and etching process. Another representative hydrogel, PVA can serve as an e-beam lithography (EBL) resist

to directly construct nanostructures with geometric shapes or various thicknesses after exposure and development.^[45] Despite that the current PVA-based nanophotonics exhibit extraordinary spectral control characteristics, their optical functionality is limited to merely near-field amplitude tuning and lacks light-field modulation capability, significantly restraining their performance in versatile, practical optical displays. Moreover, the conventional fabrication method for PVA structures usually requires multiple post-processing steps, including development and deposition after electron beam lithography exposure. Therefore, it remains an unresolved but imperative challenge to enable a hydrogel platform for dynamic multifunctional light-field tuning with simplified and fast fabrication.

Here, we originally demonstrate a direct-printing method on PVA-based nanocavity without additional post-fabrication, and harness the full potential of hydrogel nanocavity for versatile humidity-driven dynamic colorful display platforms. Through directly exposing the metal-hydrogel-metal nanocavity (MHMN) with grayscale e-beam lithography (G-EBL), the PVA layer undergoes a dose-induced thickness variation due to the polymer shrinkage, thus directly printing the functional patterns with stepwise nanometric heights. By fully exploiting the humidity-responsive, dose-induced shrinkage, and spectral selection characteristics of MHMN, we experimentally demonstrate versatile, dynamic platforms for optical encryption, dynamic colorful printing, and dynamic full-color holography displays. Specifically, the MHMN platforms can be directly

created for colorful patterns with similar or distinct transmissive colors under indoor humidity by tailoring the exposure dose. Moreover, when applying different ambient humidity, the printing colors are notably and continuously tuned in the entire visible regime with a millisecond-level response speed. More intriguingly, the directly printed PVA nanocavities with similar colors but different cavity lengths allow for transmission phase encoding at the independent wavelength channel, thus enabling the MHMN platform for humidity-driven dynamic full-color holography, beyond the fixed or monochromatic holographic switching. We envision that the direct-printed MHMN platform provides a simple and practical route for dynamic colorful optical display, and suggests a major potential for innovative nanophotonics devices with practical application in optical information security, optical sensing, and future-proof dynamic display technology.

2. Results and Discussion

Figure 1a schematically illustrates the direct grayscale-printing technology based on EBL-induced polymer shrinkage to construct MHMN with various cavity lengths and exhibit humidity-driven dynamic colorful displays. By utilizing the PVA inflation sensitivity to humidity alteration, the transmissive color of each MHMN pixel can be continuously tuned with the ambient humidity variation due to the spectral selectivity from Fabry-Perot

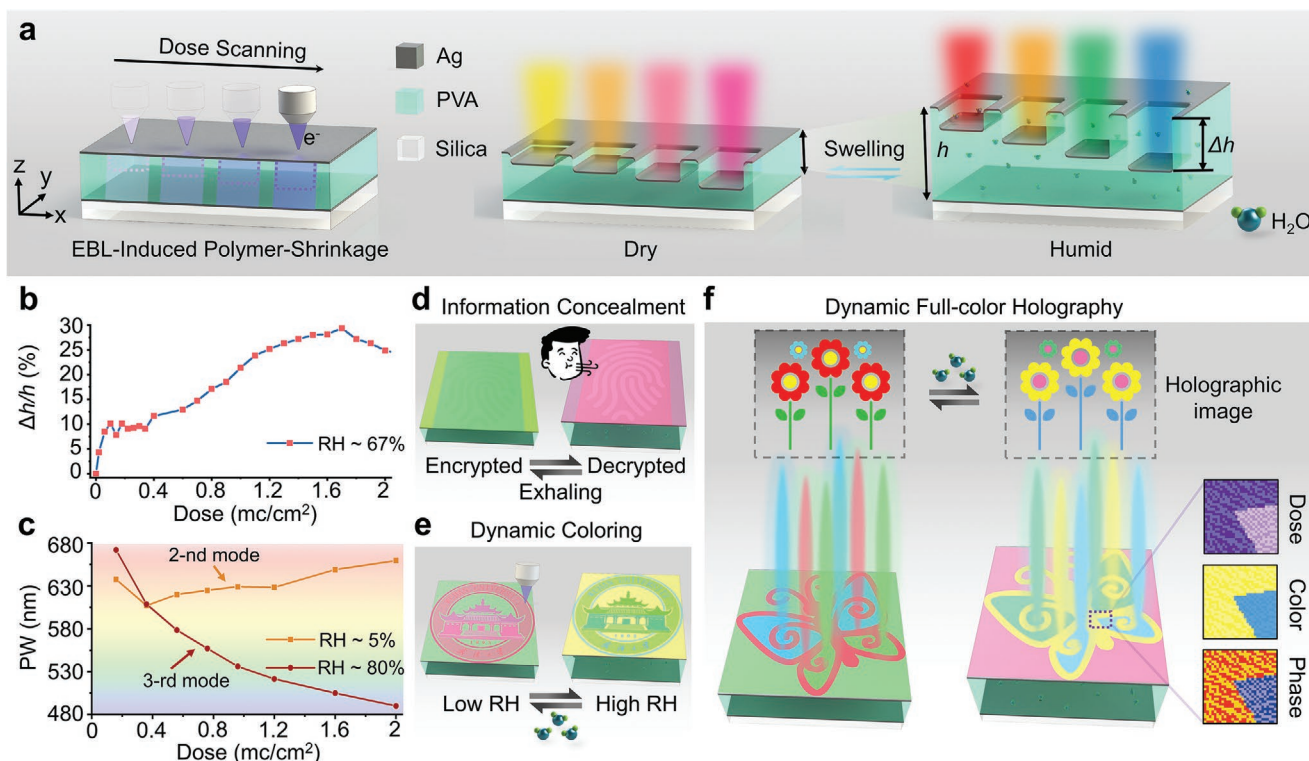


Figure 1. Direct printing hydrogel-scalable nanocavities with various thicknesses for versatile, dynamic display platforms. a) Schematic of the MHMN shrinkage under the e-beam exposure, and its swelling behavior by absorbing water molecular for the spectral tuning. b) Measured MHMN shrinkage versus exposure dose at the relative humidity (RH) of ~67%. c) The peak wavelength of the MHMN transmittance spectra is plotted as a function of the exposure dose at RH ~5% and ~80%. Graphic illustration of the humidity-responsive platforms for d) information concealment, e) dynamic coloring, and f) dynamic full-color holography by directly printing the hydrogel nanocavity.

(FP) configuration. To facilitate rapid humidity response and vivid color display, two boundary Ag layers with an ultrathin thickness of 20 nm are utilized to sandwich a ≈ 460 nm-thick PVA core layer. Because high-energy radiations could crosslink polymer chain ends without the addition of crosslinkers^[46,47] and induce the volume shrinkage,^[48] the shrinkage rate $\Delta h/h$ of MHMN architecture would increase with the exposure dose in a certain range and maxes out to $\approx 30\%$ at the dose of 1.7 mc cm^{-2} (E-beam acceleration voltage, 10 kV) under the relative humidity (RH) $\approx 67\%$, as shown in Figure 1b. The cavity shrinkage is numerically evaluated by retrieving the measured transmittance spectra from the FP etalon property, which reaches a great agreement with the profilometer measurement results (Figure S1, Supporting Information). For the humidity-driven FP resonance shift, Figure 1c shows the MHMN peak wavelength (PW) drastically transform from the 2nd order mode to the 3rd order mode when RH increases from $\approx 5\%$ to $\approx 80\%$. That is because of the excellent swelling behavior of PVA layer, and its thickness can scale up to approximately twofolds under saturated ambient humidity,^[44] thus shifting the dynamic coloring of MHMN covering the visible regime. By taking full advantage of the structural color generated by direct printing, we demonstrate the display functionality of MHMN platforms for information encrypting and dynamic coloring, as schematically shown in Figure 1d,e. In addition to the near-field exhibition, the proposed MHMN platform based on exploiting the phase modulation capability could also enable tuning the dynamic full-color multi-field display with simultaneous holography via simply regulating RH in real-time (Figure 1f).

To quantitatively study the direct-printed MHMN performance in terms of humidity-driven dynamic coloring, response/recovery speed, and repeatability, we fabricated MHMN a series of individual color plates ($30 \times 30 \mu\text{m}^2$) under various exposure doses using G-EBL (see Figure 2a). The color plates exhibit vivid and plentiful transmissive color under different humidity conditions at normal incidence with halogen lamp illumination, including magenta, orange, green, purple, pink, etc. (Video S1, Supporting Information). In contrast to the similar coloring appearance at RH $\approx 5\%$, the plate colors shift to become more distinct, covering from red to purple when the RH increases to $\approx 80\%$. This is because the PVA layer inflation amplifies the slight cavity length difference between adjacent doses. Figure 2b shows the measured transmittance of the color plates under the RH $\approx 5\%$, $\approx 40\%$, and $\approx 80\%$ (see more details in Figure S2, Supporting Information), which agrees well with the observed color change (Figure 2a). To further reveal the coloring tuning range, the corresponding dynamic color gamut for the measured spectra is mapped on the CIE 1931 Chromaticity Diagram (Figure 2c).

To investigate the effect of exposure dose on the PVA swelling performance, the cavity length change under different humidity is numerically evaluated by retrieving from the experimental transmittance spectra, as shown in Figure 2d. When the RH increases from 5% to 80%, the swelling ratio of the printed PVA thickness can reach ≈ 1.60 folds at the low exposure dose of 0.36 mc cm^{-2} , but only scales up to ≈ 1.16 folds at the high dose of 2.0 mc cm^{-2} , which is possibly attributed to the weakened hydrophilicity of PVA under the high e-beam dose exposure. To test the response/recovery speed of the MHMN, we adjust

the high humidity (RH $\approx 80\%$) surrounding the sample surface and analyze the RGB value variation of its transmissive color in real-time (Figure 2e). Right after the high humidity triggers, the color of the MHMN sample rapidly starts to transform at millisecond-level speed and undergoes a continuous shift from purple to red within 2800 ms. When the high humidity stimulus ceases, it shows a comparable recovery time to resume. Hence, it indicates an excellent capability and fast responsivity of MHMN to exchange water molecules from the environment. Regarding the sample repeatability, the cycle measurement of humidifying/drying around the sample between the RH $\approx 5\%$, 63% (indoor RH), and $\approx 80\%$ is conducted to record its coloring value variation with the inset optical image comparison (Figure 2f), which proves the long-lasting durability and repeatability of the printed patterns.

To experimentally demonstrate the feasibility of direct-printing PVA-nanocavity for optical security and dynamic coloring, we directly write on the sample with two close exposure doses to create a two-stage design for information concealment, as schematically illustrated in Figure 3a. The image of "Fingerprint" is unnoticeably concealed under the indoor humidity of $\approx 67\%$ at normal incidence. However, when increasing the surrounding humidity by simply exhaling, the hidden information becomes suddenly visible to human eyes due to the distinguished coloring difference from the PVA swelling (Figure 3b; Figure S3, Supporting Information). In addition to the repeated information retrieval, the MHMN platform can also serve for irreversible information security by utilizing the water solubility of PVA. As shown in Figure 3c, the printed QR code composed of two different heights of MHMN is readable under the indoor RH, and its color can be dynamically transformed between the RH 40% to 90% (Figure S4, Supporting Information). However, when the environment is over moisture to form water droplets, the PVA layer would quickly dissolve, leading to the irreversible destruction and removal of the QR code information. In daily life, the overly moist environment is easily accessible by deeply exhaling, allowing the conveyed information to be immediately destroyed after the recipient reads.

To visualize the dynamic printing with more color variation, we employ multiple doses to directly print a colorful pattern of the "Wuhan University badge", and its optical image is captured under scattered white light illumination (Figure 3d; Figure S5, Supporting Information). The pattern coloring is observed to exhibit a dynamic and continuous shift as RH increases from $\approx 10\%$ to $\approx 85\%$ at the response speed of millisecond-level (Video S2, Supporting Information). A minor defect in the colorful printing is sourced from the crosstalk between the adjacent pixels, which could be improved by optimizing the pixel interval (see more details in Figures S6 and S7, Supporting Information).

Furthermore, to prove the dynamic far-field holography capability of the direct-printed MHMN, we demonstrate a humidity-driven colorful printing platform with simultaneous switchable holography, as schematically illustrated in Figure 4a. By elaborately encoding the phase distribution into the printed patterns, the holographic image dynamically transforms from "Seedling" to "Flower" along with the printing "Peacock" color variation during the ambient humidity increases. Specifically, the transmitted intensity alteration of the printing

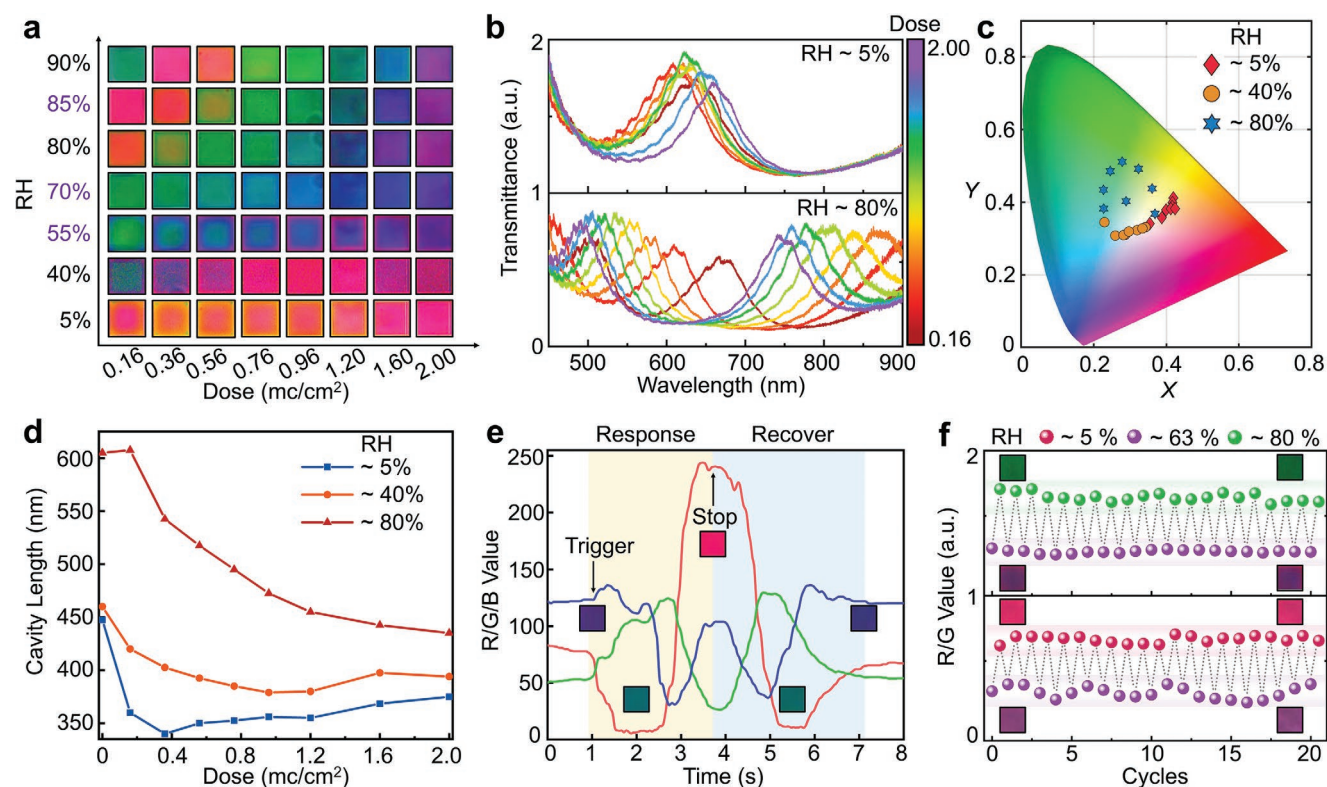


Figure 2. Optical property and humidity-response of the direct-printed MHMN. a) Optical microscopy image of the fabricated hydrogel nanocavities ($30 \times 30 \mu\text{m}^2$) with varied exposure doses at different RH. The RH value (in purple) is evaluated from the actual measured humidity value (in black). b) Measured transmittance spectra correspond to the color palettes in (a) under the RH $\approx 5\%$ and $\approx 80\%$, respectively. c) The corresponding coloring trajectory is denoted in the CIE 1931 chromaticity diagram. d) The cavity length of MHMN calculated from spectra is plotted as a function of the exposure dose under different RH. e) The response and recovery time test under high humidity stimulus. f) The repeatability test under the cycle measurements of RH adjustment.

areas (Area 1/Area 2) at the wavelength of 640 nm induces the different detour phases to be encoded into the transmitted light, thus possibly achieving the dynamic far-field holography switch. As the RH increases by exhalation, we experimentally observe that the red-dominated part converts from Area 1 to both Area 1/Area 2 coverage (Figure 4a). Figure 4b shows the captured printing color tuning as the RH increases from $\approx 50\%$ to $\approx 90\%$ (see more details in Figure S8, Supporting Information). Simultaneously, the projected holographic image under a laser illumination of 640 nm is dynamically switched from “Seedling” to “Flower” in real-time, owing to the operating pattern area alteration (see Figure 4c). Because of the broadband spectral selectivity characteristic of MHMN, the switchable holography exhibits varied tuning consequences at wavelengths of 550 and 500 nm, respectively.

To further serve as an essential technology for future-proof display, the tunable holography with full-color exhibition beyond the monochromatic on-and-off holography switch is more desirable but still challenging. For fulfilling such a meaningful functionality, we originally demonstrated the full-color dynamic holography with a synchronous dynamic colorful printing based on the MHMN platform, as shown in Figure 5a. Unlike the above detour phase design scheme, the dose-induced shrinkage empowers MHMN pixels with similar colors but various cavity thicknesses, thus allowing the pixels

to exhibit the transmission phase modulation at the selected wavelength (Figure 5b). Figure 5c shows the measured transmittance comparison of Areas 1 and 2 under the indoor RH $\approx 60\%$ at the wavelength of 480 and 550 nm, which implies the favorable intensity contrast for independent dual-color holography (Figure S9, Supporting Information). As a proof-of-concept demonstration, we directly print an “Owl” pattern in the area of $750 \times 750 \mu\text{m}^2$ (500×500 pixels) with elaborate-selected exposure doses. Figure 5d shows the continuous dynamic coloring of the printed pattern during the RH increasing process, which exhibits rapid and uniform response at the millisecond-level. More interestingly, the color of the far-field holographic image “Puzzle” transforms from blue/green (B/G) to green/red (G/R) by simply exhaling, and shows a good contrast at the RGB wavelength channels (Figure 5e). Note that to ensure the image quality, the holographic image is measured under the laser illumination wavelengths of 480/550 nm or 550/650 nm, respectively (see more details in Figure S10, Supporting Information). Experimental animation to capture the full-color holographic dynamics in real-time can be found in Video S3 (Supporting Information).

To expand the coloring capability of dynamic holography, we further generate extra holographic color based on the color-mixed scheme^[49] (Figure 6a). Figure 6b shows the design framework of the target holographic image optimization for

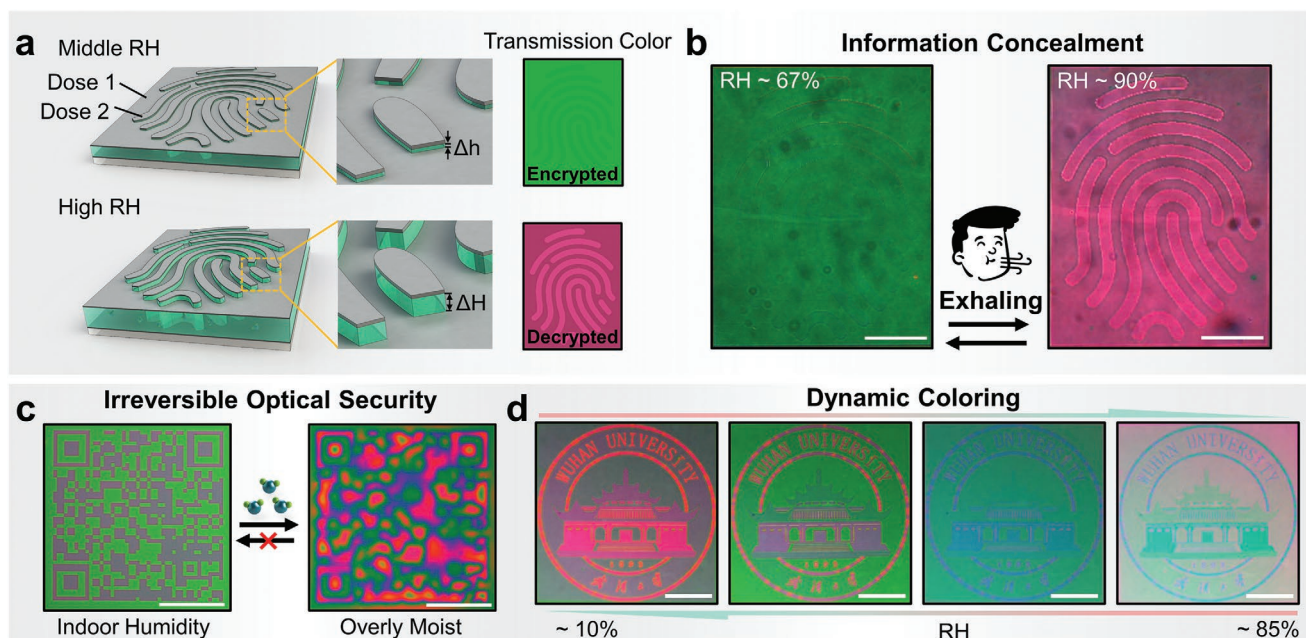


Figure 3. Direct-printing structural color for humidity-responsive optical security and dynamic display. a) Schematic illustration of the MHMN fabrication for a hidden pattern “Fingerprint” and its working mechanism of humidity-triggered decryption. b) Experimentally captured optical image of the “Fingerprint” pattern under the indoor RH $\approx 67\%$ and high RH $\approx 90\%$ from exhalation. The scale bar is $100\ \mu\text{m}$. c) Optical micrograph comparison of the fabricated “QR code” pattern for prior-/post-exposure to overly moist conditions. The scale bar is $100\ \mu\text{m}$. d) Optical images of the “Wuhan University badge” pattern under multi-level humidity tuning. The scale bar is $200\ \mu\text{m}$.

color mixing, which adopts two primary colors (green and red) to overlap and generate the yellow letter “C”. After RGB component extraction and compensation processing, the phase distribution for the optimized target image is calculated and encoded into the printing contour using the simulated annealing algorithm (SAA). Due to the two-step phase design, the holographic image would be accompanied by a conjugate image

that is symmetric about the zero-order. By employing a dual-color pattern “Forest” of $750 \times 750\ \mu\text{m}^2$ area (500×500 pixels) as the color printing, the projected holographic image can be engineered to degenerate and split to exhibit different images at the wavelength channel of 480 and 550 nm (at the indoor RH $\approx 60\%$) due to spectral selection (Figure S11, Supporting Information). When the RH is adjusted to $\approx 90\%$, the printing colors

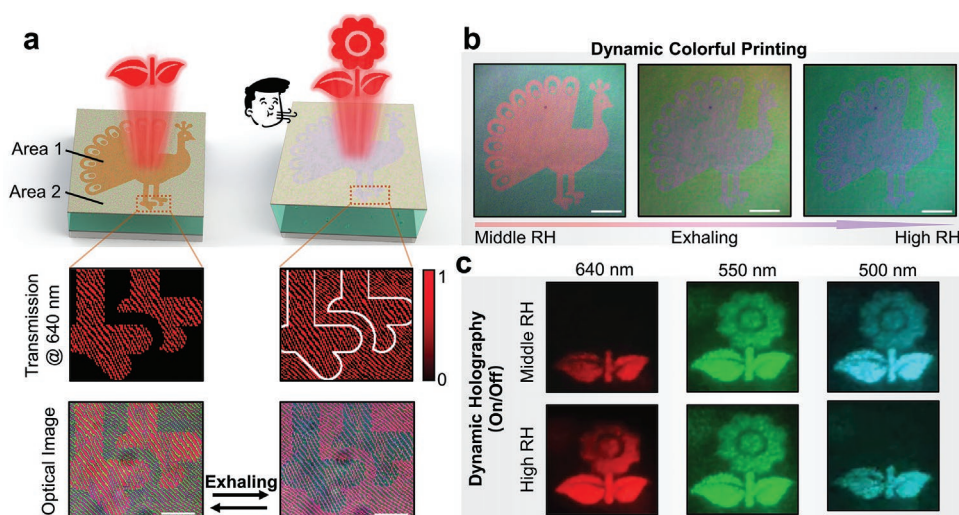


Figure 4. Dynamic holography display triggered simply by exhalation. a) Schematic diagram of the tuning mechanism for dynamic far-field holographic projection from simultaneous structural colorful printing. The scale bar is $50\ \mu\text{m}$. b) The color variation of the printing “Peacock” captured by microscope during the humidity increase. The scale bar is $200\ \mu\text{m}$. c) Experimentally observed holography display dynamics during the exhalation process under the laser illumination wavelength of 640, 550, and 500 nm, respectively.

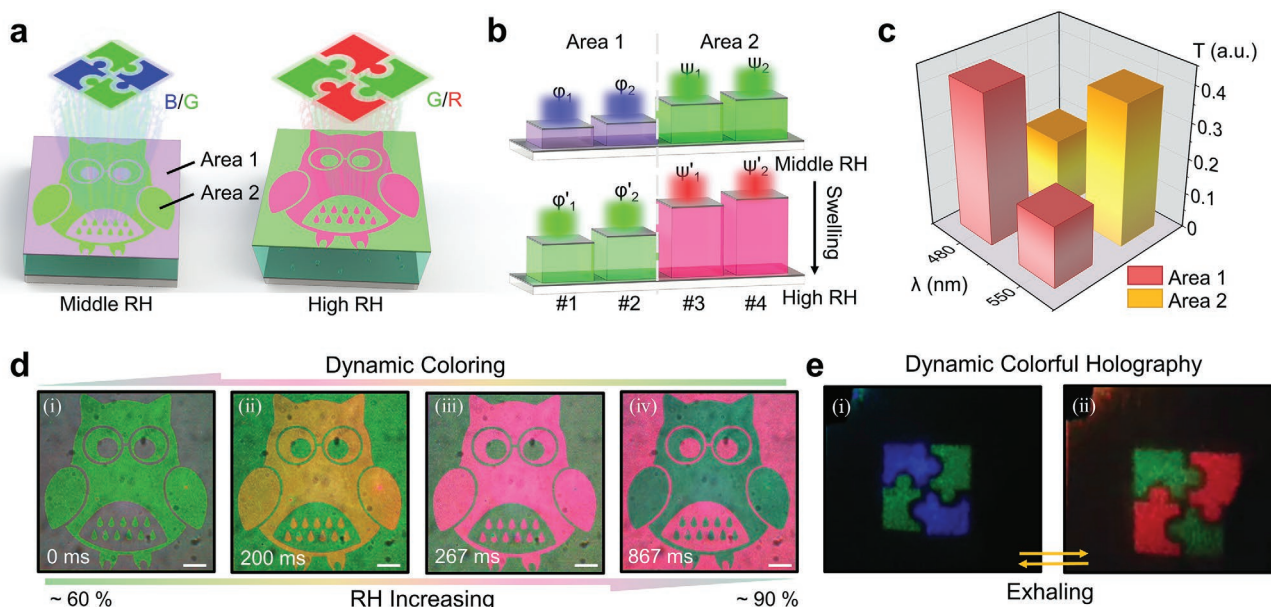


Figure 5. Dynamic full-color holography. a) Schematic of the colorful holographic image “Puzzle” transforms from B/G to G/R due to the humidity alternation, simultaneously accompanied with the color shift of printing image “Owl”. b) Four coloring pixels generated by various exposure doses to construct the dual-color printing and provide a two-phase choice for encoding the colorful holography. c) Measured transmittance comparison for patterns Area 1 and Area 2 at the respective wavelength of 480 and 550 nm under the indoor RH $\approx 60\%$. d) The printing image triggered by RH increase exhibits a continuous color shift at millisecond-level response speed. The scale bar is 100 μm . e) Observed transformation of the dynamic full-color holography during the exhalation process at the wavelength channels of i) 480/550 nm and ii) 550/650 nm.

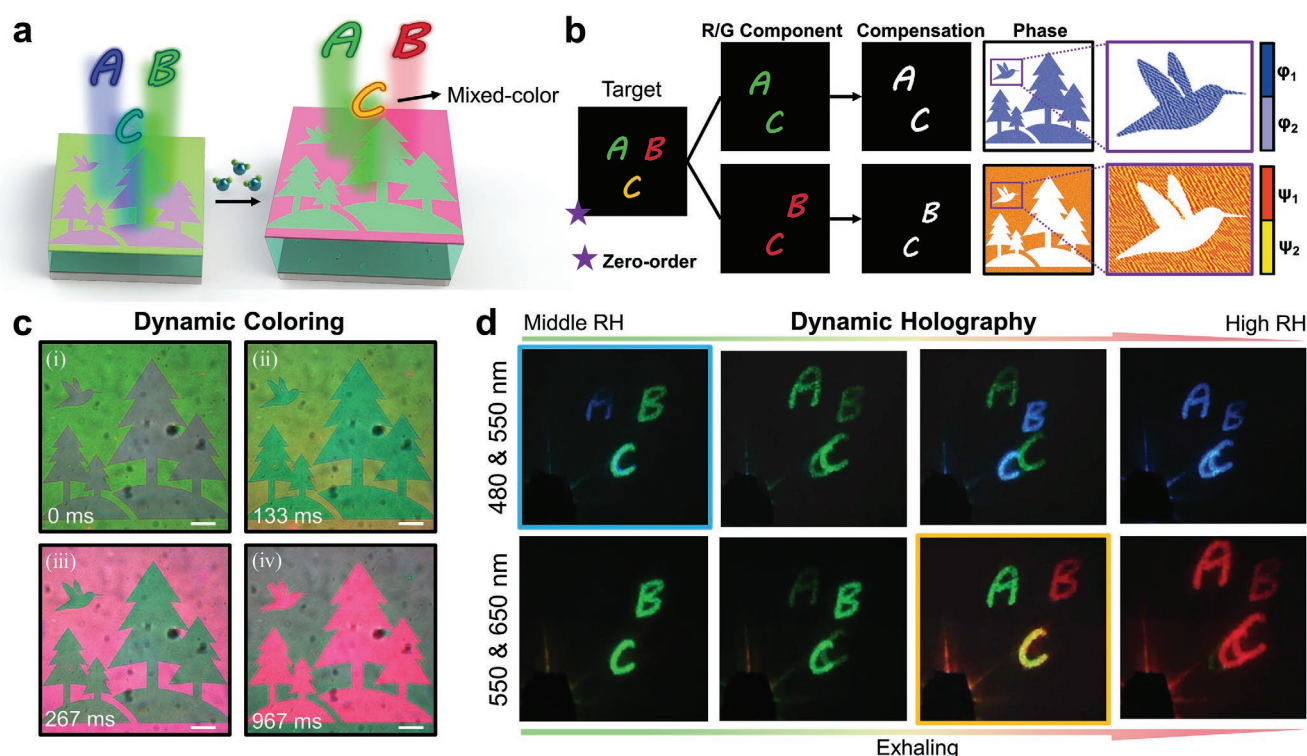


Figure 6. Dynamic color-mixed holography. a) Schematic diagram for the dynamic full-color holographic display to generate a color-mixed letter of yellow “C”. b) The design framework for color-mixed holography projection from the simultaneous colorful printing. The holographic image is designed off-axis to avoid zero-order diffraction. c) Coloring dynamics of the printing “Forest” after exposure to high humidity. The scale bar is 100 μm . d) The dynamic color-mixed holographic images are experimentally captured during the exhalation process.

drastically change as expected (Figure 6c), and meanwhile its accompanied holographic color also exhibits plentiful color variations (Figure 6d).

Regarding the tuning rate, the color of the printed “Forest” begins to change within 150 ms and reaches a maximum shift at ≈ 1000 ms after exposure to high humidity. Due to the wavelength dispersion relation, we experimentally observe a cyan holographic letter “C” at the B/G illumination channels under indoor RH, however, a yellow letter “C” is observed at the G/R illumination channels during the humidification process (Video S4, Supporting Information). The proposed MHMN platform can implement more complex dynamic full-color holography by involving more colored pixels in the design. Moreover, it is anticipated that the adoption of a precise humidity control system can further ensure the practical application scenarios of the hydrogel direct-printed technique for dynamic animation display functionalities.

3. Conclusion

In summary, we demonstrate a humidity-driven MHMN platform using direct-printing technology for versatile, dynamic colorful displays, including optical encryption, dynamic multicolor printing, and dynamic full-color holography. The dose-induced PVA shrinkage under e-beam exposure is originally investigated to directly generate stepwise MHMN with tunable structural color. Through employing the ambient humidity as the stimulus, the direct-printed MHMN exhibit vivid dynamic coloration, and rapid millisecond-level response. Therefore, by fully exploiting the tunable optical property of the hydrogel-based nanocavity, we experimentally demonstrate the optical concealment, dynamic coloring, and switchable holography functionalities. Moreover, by utilizing the cavity-dependent transmission phase, the MHMN platform can also enable dynamic full-color holography in real-time with color-mixing capability, for the first time, beyond the monochromatic holography switching. Overall, we believe that such a direct-printed hydrogel-tunable platform featured post-fabrication free, repeatable/rapid humidity-responsivity, great simplicity of tuning method, large-area modulation, and full-color potential, provides a new paradigm for dynamic meta-optics display and suggest practical applications including optical security, gas sensing, multispectral imaging, and future-proof display.

4. Experimental Section

Numerical Simulations: The finite-difference time-domain (FDTD) method was utilized to calculate the transmittance spectra of the MHMN with the PVA layer thickness variation. The complex refractive index of Ag for simulation was utilized from the data of Palik ($0\text{--}2\text{ }\mu\text{m}$). The MHMN architecture was modeled as a unit cell of $20 \times 20\text{ nm}^2$. Perfectly matched layers (PML) were utilized along the propagation of waves (z-axis), and periodic boundary conditions along the x-axis and y-axis. Regarding the cavity length evaluation, its refractive index was assumed to remain at a constant 1.51 in the numerical calculations, disregarding the minor index-alteration impact.^[50]

Sample Fabrications: For the MHMN platform preparation, Ag/PVA/Ag films were successively deposited on a silica substrate. For the hydrogel layer, the 10.7 wt% PVA solution was spin-coated onto the sample at the

speed of 4000 rpm for 60 s to give a ≈ 460 nm thick hydrogel layer. The root mean square (RMS) roughness of the coated hydrogel film was characterized to < 0.4 nm by utilizing atomic force microscopy (AFM) measurement (see Figure S13, Supporting Information). The thick Ag films were deposited by thermal evaporation (pressure of 1×10^{-4} Pa and evaporation rate of $0.2\text{--}0.3\text{ }\text{\AA}\text{ s}^{-1}$), and the RMS roughness of the top Ag film was < 1.6 nm. For the direct printing on the MHMN, the sample was directly exposed to G-EBL (Raith 150, electron acceleration voltage of 10 kV).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

The authors gratefully acknowledge the National Key Research and Development Program of China (no. 2022YFB3808600), National Natural Science Foundation of China (no.12204360), Hubei Province Funds for Distinguished Young Scientists (2021CFA043), Wuhan Science and Technology Bureau (2020010601012196), and the Natural Science Foundation of Jiangsu Province (BK20220281). The authors thank the Core Facility of Wuhan University for the measurements of AFM. The work was supported by the Center for NanoScience and Nanotechnology at Wuhan University, the Large-scale Instrument and Equipment Sharing Foundation of Wuhan University, and the Hubei Luojia Laboratory.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

direct printing, dynamic displays, full-color holography, humidity-responsive behavior, hydrogel platforms

Received: October 18, 2022

Revised: November 25, 2022

Published online:

- [1] D. Lin, P. Fan, E. Hasman, M. L. Brongersma, *Science* **2014**, *345*, 298.
- [2] A. Arbabi, E. Arbabi, Y. Horie, S. M. Kamali, A. Faraon, *Nat. Photonics* **2017**, *11*, 415.
- [3] Q. Song, M. Odeh, J. Zúñiga-Pérez, B. Kanté, P. Genevet, *Science* **2021**, *373*, 1133.
- [4] R. Zheng, R. Pan, G. Geng, Q. Jiang, S. Du, L. Huang, C. Gu, J. Li, *Nat. Commun.* **2022**, *13*, 4292.
- [5] F. Neubrech, X. Duan, N. Liu, *Sci. Adv.* **2020**, *6*, eabc2709.
- [6] X. Cao, Y. Du, Y. Guo, G. Hu, M. Zhang, L. Wang, J. Zhou, Q. Gao, P. Fischer, J. Wang, *Adv. Mater.* **2022**, *34*, 2109161.

- [7] S. Daqiqeh Rezaei, J. Ho, T. Wang, S. Ramakrishna, J. K. Yang, *Nano Lett.* **2020**, 20, 4422.
- [8] P. Dai, K. Sun, X. Yan, O. L. Muskens, C. K. de Groot, X. Zhu, Y. Hu, H. Duan, R. Huang, *Nanophotonics* **2022**, 11, 3057.
- [9] N. Yu, P. Genevet, M. A. Kats, F. Aieta, J.-P. Tetienne, F. Capasso, Z. Gaburro, *Science* **2011**, 334, 333.
- [10] Z. Wang, C. Dai, Z. Li, Z. Li, *Nano Lett.* **2022**, 22, 2059.
- [11] Z. Li, C. Wan, C. Dai, J. Zhang, G. Zheng, Z. Li, *Adv. Opt. Mater.* **2021**, 9, 2100297.
- [12] Y. Ni, S. Chen, Y. Wang, Q. Tan, S. Xiao, Y. Yang, *Nano Lett.* **2020**, 20, 6719.
- [13] G. Zheng, H. Mühlenbernd, M. Kenney, G. Li, T. Zentgraf, S. Zhang, *Nat. Nanotechnol.* **2015**, 10, 308.
- [14] Y. Shi, C. Wan, C. Dai, S. Wan, Y. Liu, C. Zhang, Z. Li, *Optica* **2022**, 9, 670.
- [15] R. Yang, S. Wan, Y. Shi, Z. Wang, J. Tang, Z. Li, *Laser Photonics Rev.* **2022**, 16, 2200127.
- [16] S. Zhang, L. Huang, G. Geng, J. Li, X. Li, Y. Wang, *Photonics Res.* **2022**, 10, 1031.
- [17] H. Yang, Y. Jiang, Y. Hu, K. Ou, H. Duan, *Laser Photonics Rev.* **2022**, 16, 2200351.
- [18] Z. Li, C. Wan, C. Dai, Z. Li, *Small* **2022**, 2205041.
- [19] M. Khorasaninejad, W. T. Chen, R. C. Devlin, J. Oh, A. Y. Zhu, F. Capasso, *Science* **2016**, 352, 1190.
- [20] J. Li, Y. Wang, C. Chen, R. Fu, Z. Zhou, Z. Li, G. Zheng, S. Yu, C. W. Qiu, S. Zhang, *Adv. Mater.* **2021**, 33, 2007507.
- [21] Y. Wang, Q. Chen, W. Yang, Z. Ji, L. Jin, X. Ma, Q. Song, A. Boltasseva, J. Han, V. M. Shalae, S. Xiao, *Nat. Commun.* **2021**, 12, 5560.
- [22] C. Wan, Z. Li, S. Wan, C. Dai, J. Tang, Y. Shi, Z. Li, *Adv. Funct. Mater.* **2022**, 32, 2110592.
- [23] I. Kim, M. A. Ansari, M. Q. Mehmood, W. S. Kim, J. Jang, M. Zubair, Y. K. Kim, J. Rho, *Adv. Mater.* **2020**, 32, 2004664.
- [24] J. Wang, K. Li, H. He, W. Cai, J. Liu, Z. Yin, Q. Mu, V. K. Hisao, D. Gérard, D. Luo, *Laser Photonics Rev.* **2022**, 16, 2100396.
- [25] K. Li, J. Wang, W. Cai, H. He, M. Cen, J. Liu, D. Luo, Q. Mu, D. Gérard, Y. J. Liu, *Nano Lett.* **2021**, 21, 7183.
- [26] J. Liu, H. Zeng, M. Cheng, Z. Wang, J. Wang, M. Cen, D. Luo, A. Priimagi, Y. J. Liu, *Mater. Horiz.* **2022**, 9, 942.
- [27] K. Li, J. Wang, W. Cai, H. He, J. Liu, Z. Yin, D. Luo, Q. Mu, D. Gérard, Y. J. Liu, *Opt. Express* **2022**, 30, 31913.
- [28] H. He, J. Wang, K. Li, M. Cen, W. Cai, J. Liu, D. Luo, Y. Q. Lu, Y. J. Liu, *Laser Photonics Rev.* **2022**, 16, 2200115.
- [29] S. Abdollahramezani, O. Hemmatyar, M. Taghinejad, H. Taghinejad, A. Krasnok, A. A. Eftekhar, C. Teichrib, S. Deshmukh, M. A. El-Sayed, E. Pop, *Nat. Commun.* **2022**, 13, 1696.
- [30] Y. Zhang, C. Fowler, J. Liang, B. Azhar, M. Y. Shalaginov, S. Deckoff-Jones, S. An, J. B. Chou, C. M. Roberts, V. Liberman, *Nat. Nanotechnol.* **2021**, 16, 661.
- [31] C. R. de Galarreta, I. Sinev, A. M. Alexeev, P. Trofimov, K. Ladutenko, S. G.-C. Carrillo, E. Gemo, A. Baldycheva, J. Bertolotti, C. D. Wright, *Optica* **2020**, 7, 476.
- [32] S. Chen, S. Rossi, R. Shanker, G. Cincotti, S. Gamage, P. Kühne, V. Stanishev, I. Engquist, M. Berggren, J. Edberg, *Adv. Mater.* **2021**, 33, 2102451.
- [33] Y.-W. Huang, H. W. H. Lee, R. Sokhoyan, R. A. Pala, K. Thyagarajan, S. Han, D. P. Tsai, H. A. Atwater, *Nano Lett.* **2016**, 16, 5319.
- [34] S. M. Kamali, E. Arbabi, A. Arbabi, Y. Horie, A. Faraon, *Laser Photonics Rev.* **2016**, 10, 1002.
- [35] X. Zhang, Y. Zhou, H. Zheng, A. E. Linares, F. C. Ugwu, D. Li, H.-B. Sun, B. Bai, J. G. Valentine, *Nano Lett.* **2021**, 21, 8715.
- [36] C. Zhang, J. Jing, Y. Wu, Y. Fan, W. Yang, S. Wang, Q. Song, S. Xiao, *ACS Nano* **2019**, 14, 1418.
- [37] M. R. Shcherbakov, S. Liu, V. V. Zubyuk, A. Vaskin, P. P. Vabishchevich, G. Keeler, T. Pertsch, T. V. Dolgova, I. Staude, I. Brener, *Nat. Commun.* **2017**, 8, 17.
- [38] I. Kim, J. Yun, T. Badloe, H. Park, T. Seo, Y. Yang, J. Kim, Y. Chung, J. Rho, *Photonics Res.* **2020**, 8, 1409.
- [39] J. Li, Y. Chen, Y. Hu, H. Duan, N. Liu, *ACS Nano* **2020**, 14, 7892.
- [40] X. Cui, X. Zhu, L. Shao, J. Wang, A. Kristensen, *Adv. Opt. Mater.* **2020**, 8, 1901605.
- [41] Y. Zhang, Q. Zhang, X. Ouyang, D. Y. Lei, A. P. Zhang, H.-Y. Tam, *ACS Nano* **2018**, 12, 9913.
- [42] J. Zhang, D. Wang, Y. Ying, H. Zhou, X. Liu, X. Hu, Y. Chen, Q. Li, X. Zhang, M. Qiu, *Nanophotonics* **2021**, 10, 4125.
- [43] C. Jung, S.-J. Kim, J. Jang, J. H. Ko, D. Kim, B. Ko, Y. M. Song, S.-H. Hong, J. Rho, *Sci. Adv.* **2022**, 8, eabm8598.
- [44] Z. Wang, C. Dai, J. Zhang, D. Wang, Y. Shi, X. Wang, G. Zheng, X. Zhang, Z. Li, *Adv. Funct. Mater.* **2022**, 32, 2110022.
- [45] J. Zhang, C. Huang, Y. Chen, H. Wang, Z. Gong, W. Chen, H. Ge, X. Hu, X. Zhang, *Nanotechnology* **2020**, 31, 425303.
- [46] Y. Hu, L. Li, Y. Wang, M. Meng, L. Jin, X. Luo, Y. Chen, X. Li, S. Xiao, H. Wang, Y. Luo, C. Qiu, H. Duan, *Nano Lett.* **2019**, 20, 994.
- [47] N. A. Peppas, E. W. Merrill, *J. Appl. Polym. Sci.* **1977**, 21, 1763.
- [48] G. Tzvetkov, P. Fernandes, S. Wenzel, A. Fery, G. Paradossi, R. H. Fink, *Phys. Chem. Chem. Phys.* **2009**, 11, 1098.
- [49] Y. Hu, L. Li, Y. Wang, M. Meng, L. Jin, X. Luo, Y. Chen, X. Li, S. Xiao, H. Wang, *Nano Lett.* **2019**, 20, 994.
- [50] J. Zhang, Q. Li, C. Dai, M. Cheng, X. Hu, H.-S. Kim, H. Yang, D. Preston, Z. Li, X. Zhang, *Small* **2022**, 18, 202205057.